SAP ABAP PROJECT REPORT

RealABAPDevelopmentScenario

Custom Sales Order ALV Report (ZSOLIST)

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Project Topic	Real ABAP Development Scenario — Custom ALV Report (Topic vi)
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1. Project Title

CustomSalesOrderALV Report in SAP ABAP (Report: ZSOLIST)

2. Problem Statement

In large organizations utilizing SAP ERP, the Sales and Distribution (SD) module processes numerous sales orders per day for different sales organizations, different distribution channels, and different divisions. Users often require an overview of the sales orders that have been created and processed by their respective organizations, along with the line items of the orders containing material, quantity, unit of measure, net value, and currency. The standard SAP transaction VA05 (List of Sales Orders) gives a general listing but
is unable to provide flexibility in terms of displaying organization-based information and field catalogs. Also, the reporting function of the SD module does not always provide an integrated header-and-item flat view which could be filtered and sorted.
The aim of this project is to create a custom report in ABAP (ZSOLIST) using the SAP ALV Grid that will fetch data from the VBAK (Sales Order Header) and VBAP (Sales Order Item) tables and then show the output in an interactive ALV grid format with zebra row style and auto optimized columns.

3. Solution / Features

3.1 Technical Architecture

The solutionisaclassicalABAP report that leverages the Object-Oriented ALV Grid control (CL_GUI_ALV_GRID) embedded inside a custom container (CL_GUI_CUSTOM_CONTAINER) rendered on a custom Dynpro screen (Screen 100). The application follows a clean three-layer pattern:

- Data Layer: A single SELECT with an INNER JOIN on VBAK and VBAP retrieves the relevant fields (up to 100 rows) into a transparent custom structure ZSSODATA.
- Presentation Layer: The ALV grid is instantiated as an object, bound to the container on Screen 100, and displayed using SET_TABLE_FOR_FIRST_DISPLAY.
- Event Layer: User commands (BACK, EXIT, CANCEL) are handled via the MODULE USER_COMMAND_0100 to ensure clean navigation.

3.2 Key Features

- Combined Headers & Items Data Retrieval: Retrieves VBELN (sales order), POSNR (item), AUART (order type), VKORG (sales organization), VTWEG (channel distribution), SPART (division), MATNR (material), PSTYV (item category), ZMENG (target quantity), ZIEME (measurement unit), NETWR (net value), and WAERK (currency) all in one go via SQL.
- Grid Control Using CL_GUI_ALV_GRID: Uses the updated ALV control class that offers sort, filter, column resize, spreadsheet export, and print preview functionality right out of the box.
- Zebra Lines (gs_layout-zebra): Activates zebra lines to enhance readably.
- Automatic Columns Optimization (gs_layout-cwidth_opt): Automatically optimizes the width of columns, saving developers' time spent creating a field catalog manually.
- Structure-Based Field Catalog: The field catalog is generated automatically based on the custom Dictionary structure of ZSSODATA.
- Specific PF Status & Title Bar Customization: In screen 100, we set a specific PF-status (SOLIST) and title (SOL) bars for better user experience.
- Limit to 100 Rows: The UP TO 100 ROWS clause helps avoid performance issues during demonstrations and unit tests, while the technique works by excluding this clause in production.

4. Technology Stack

Component	Details
SAP Platform Programming	SAP ERP / SAP S/4HANA (on-premise or trial landscape)
Language ALV Framework	ABAP (Advanced Business Application Programming)
Container Class Database	CL_GUI_ALV_GRID — Object-Oriented ALV Grid Control
Tables Custom Dictionary	CL_GUI_CUSTOM_CONTAINER
Object Screen Technology	VBAK (Sales Order Header), VBAP (Sales Order Item)
GUI Status / Title SQL	ZSSODATA (transparent structure for ALV output)
Variant Development	SAP Dynpro — Screen 100, Module Pool programming
Environment	PF-Status: SOLIST | Title Bar: SOL
	Open SQL with INNER JOIN and UP TO N ROWS clause
	SAP ABAP Workbench / Eclipse ADT (SE38 / SE80)

5. End-to-End Development Process

The following sectiondescribesthecomplete step-by-step process followed to build and activate the ZSOLIST custom ALV report in the SAP system.

Step 1: Create Custom Dictionary Structure (ZSSODATA)

Transaction: SE11 (ABAPDictionary)

1. Open SE11, select'Data type', enter name ZSSODATA, click Create.
2. Select 'Structure'asthe category.
3. Add the followingfields with their corresponding data elements:

#	Field Name	Data Element / Domain
1	VBELN	VBELN_VA — Sales and Distribution Document Number
2	POSNR	POSNR_VA — Item Number of the SD Document
3	AUART	AUART — Sales Document Type
4	VKORG	VKORG — Sales Organization
5	VTWEG	VTWEG — Distribution Channel
6	SPART	SPART — Division
7	MATNR	MATNR — Material Number
8	PSTYV	PSTYV — Sales Document Item Category
9	ZMENG	ZMENG — Target Quantity in Sales Units
10	ZIEME	VRKME — Sales Unit of Measure
11	NETWR	NETWR — Net Value of the Order Item
12	WAERK	WAERK_D — Document Currency

4. Activate thestructure (Ctrl+F3).

Step 2: Create the Custom Screen (Dynpro 100)

Transaction: SE80 or SE51

5. Open the report program ZSOLIST in SE80.
6. Create a new screen with number 100.
7. In the Layout editor, place a Custom Control named SOLIST covering the full screen area.
8. In the Flow Logic of Screen 100, add the following:

```
PROCESS BEFOREOUTPUT.
MODULE status_0100.

PROCESS AFTER INPUT.
```

```
MODULE user_command_0100.
```

9. Save and activate Screen 100.

Step 3: Create PF-Status and Title Bar

Transaction: SE41 (Menu Painter) or directly via SE80

10. In the program ZSOLIST, go to Goto > GUI Status.
11. Create GUI Status with name 'SOLIST'. Assign standard ALV function keys: BACK (F3), EXIT (Shift+F3), CANC (F12).
12. Create Title Bar with name 'SOL' and set the text to 'Sales Order List'.
13. Activate both objects.

Step 4: Write and Activate the ABAP Report (ZSOLIST)

Transaction: SE38

14. Create a new report ZSOLIST (type: Executable Program, No Selection Screen required for basic version).
15. Enter the complete ABAP code as provided in Section 6 of this document.
16. Perform a syntax check (Ctrl+F2) — ensure no errors.
17. Activate the program (Ctrl+F3).

Step 5: Execute and Test

Transaction: SE38 → Execute (F8) or assign a transaction code via SE93.

18. Run the report using F8 from SE38.
19. Verify that Screen 100 opens and the ALV grid displays sales order data.
20. Confirm zebra-row formatting is applied (alternating row colors).
21. Confirm column widths are auto-optimized.
22. Test navigation: press BACK / EXIT / CANCEL to ensure the screen closes cleanly.
23. Test ALV standard functions: sorting by VBELN, filtering by VKORG, exporting to local file.

6. ABAP Source Code — Report ZSOLIST

The complete source codeforthecustom ALV report ZSOLIST is provided below. This code is also submitted as a separate .txt file (Module_12_-_SAP_ABAP_Report_Code.txt) as part of the project ZIP package.

Program Code:

```
*&---------------------------------------------------------------------
-*
*& Report ZSOLIST
*& Custom Sales Order ALV Grid Report
*&---------------------------------------------------------------------
-*
REPORT zsolist.

DATA: gt_sod      TYPE STANDARD TABLE OF zssodata,
      o_container TYPE REF TO cl_gui_custom_container,
      o_grid      TYPE REF TO cl_gui_alv_grid,
      gs_layout   TYPE lvc_s_layo.

* Fetch Sales Order Header + Item data
SELECT a~vbeln,
b~posnr,
a~auart,
a~vkorg,
a~vtweg,
a~spart,
b~matnr,
b~pstyv,
b~zmeng,
b~zieme,
b~netwr,
b~waerk
   UP TO 100 ROWS
   INTO TABLE @gt_sod
   FROM vbak AS a
  INNER JOIN vbap AS b
     ON b~vbeln EQ a~vbeln .

* ALV Layout Settings
gs_layout-zebra      = abap_true .
gs_layout-cwidth_opt = abap_true .

*      Create      Custom
Container  CREATE  OBJECT
o_container  EXPORTING
    container_name             ='SOLIST'
 EXCEPTIONS
```

```
    cntl_error                  = 1
    cntl_system_error           = 2
    create_error                = 3
    lifetime_error              = 4
    lifetime_dynpro_dynpro_link = 5
    OTHERS                      = 6.
IF sy-subrc <> 0.
ENDIF.

* Create ALV Grid Object
CREATE OBJECT o_grid
  EXPORTING
    i_parent          = o_container
  EXCEPTIONS
    error_cntl_create  =  1
    error_cntl_init    =  2
    error_cntl_link    =  3
    error_dp_create    =  4
    OTHERS             =  5.
IF sy-subrc <> 0.
ENDIF.

* Display ALV Grid
CALL METHOD o_grid->set_table_for_first_display
  EXPORTING
    i_structure_name              = 'ZSSODATA'
    is_layout                     = gs_layout
  CHANGING                        = gt_sod
    it_outtab
  EXCEPTIONS
    invalid_parameter_combination  =  1
    program_error                  =  2
    too_many_lines                 =  3
    OTHERS                         =  4
IF sy-subrc <> 0.
ENDIF.

CALL SCREEN 100 .

*&---------------------------------------------------------------------
-*
*& Module STATUS_0100 OUTPUT
*&---------------------------------------------------------------------
-*
MODULE status_0100 OUTPUT.
  SET PF-STATUS 'SOLIST'.
  SET TITLEBAR 'SOL'.
ENDMODULE.
```

```
*&---------------------------------------------------------------------
-*
*& Module USER_COMMAND_0100 INPUT
*&---------------------------------------------------------------------
-*
MODULE user_command_0100 INPUT.
  CASE sy-ucomm .
    WHEN 'BACK' OR 'EXIT' OR 'CANC' .
      SET SCREEN 0 .
      LEAVE SCREEN .
  ENDCASE .
ENDMODULE.
```

7. Unique Points / Value Additions

- SQL JOIN in One PASS: By performing an SQL JOIN between the VBAK and VBAP tables using one SELECT statement, this report has been optimized for avoiding nested SELECT statements, which is considered an ABAP anti-pattern that significantly reduces efficiency.
- IT_FIELDCATALOG Through Dictionary: This technique does not hard-code all columns in an IT_FIELDCATALOG. Instead, this program uses I_STRUCTURE_NAME = 'ZSSODATA' and SAP determines everything else from the dictionary.
- OO-ALV vs. Classic ALV: The report uses the CL_GUI_ALV_GRID object-oriented class rather than the traditional function module REUSE_ALV_GRID_DISPLAY. Object Oriented ALV offers better flexibility, allows event handling, and is in line with the current programming principles by SAP.
- LVC_S_LAYO Layout Type: The use of the LVC_S_LAYO layout structure (instead of SLIS_T_LAYOUT_ALV for classic ALV) is accurate since this is the correct way to implement the OO-ALV grid.
- Modular Screen Structure: The separation of program logic into STATUS_0100 (output module) and USER_COMMAND_0100 (input module) adheres to the Dynpro modularization guidelines set by SAP, ensuring future scalability in implementing additional parameters to the selection screen.

8. Future Improvements

- Selection Screen: Introduce a block under PARAMETER ID or SELECT-OPTIONS that allows the filtering of the report using Sales Organization, Document Type, and Date Range.
- Field Catalog: Create IT_FIELDCATALOG through manual coding to enable field sequencing, elimination of technical fields, use of reference field for Currency/Quantity, and tooltips.
- ALV Events - Double Click: Program the DOUBLE_CLICK event of CL_GUI_ALV_GRID so that double clicking the line triggers an event that leads the program to transaction VA03 (Sales Order).
- Variant Management: Allow the use of ALV variants by implementing IS_VARIANT and I_SAVE = 'A'.
- Sorting and Subtotals: Include IT_SORT and subtotals of NETWR by VKORG to provide a financial snapshot of each Sales organization for the managers.
- Authorization Check: Incorporate AUTHORITY-CHECK on object V_VBAK_VKO (Sales Organization) prior to the SELECT statement to achieve access data control consistent with the SAP authorization model.

- Unit Tests (ABAP Unit): Write local test classes using the ABAP Unit framework to validate SQL result counts, layout flag settings, and screen flow — aligning with clean ABAP and TDD practices.
- Fiori Integration: In an S/4HANA landscape, wrap the business logic in a CDS (Core Data Service) view and expose it via OData to build a Fiori Elements list report — replacing the classical Dynpro UI with a modern, responsive interface.

9. SAP Object

AllSAPdevelopmentobjectsthat must be created and activated before executing thereport:

#	Object Name	Object Type	Transaction
1	ZSSODATA	Dictionary Structure	SE11
2	ZSOLIST	ABAP Report (Executable Program)	SE38 / SE80
3	Screen 100	Dynpro Screen (Module Pool)	SE51 / SE80
4	SOLIST	GUI Status (PF-Status)	SE41 / SE80
5	SOL	Title Bar	SE41 / SE80

10. Conclusion

TheZSOLIST custom ALV report demonstrates a complete, real-world ABAP development scenario that integrates SAP Dictionary design, Open SQL performance best practices, Object-Oriented ALV Grid programming, and Dynpro screen architecture. The report fulfills a genuine business requirement — giving sales teams an efficient, interactive, and exportable view of sales order data — while showcasing the core competencies expected of an SAP ABAP developer.

The project is built entirely from scratch as an individual effort, following the guidelines provided in the KIIT SAP Project Work Guidance document and the Capstone Project Submission Guidelines. All SAP development objects (structure, report program, Dynpro screen, GUI status, and title bar) are documented and the source code is submitted as a separate file alongside this report.